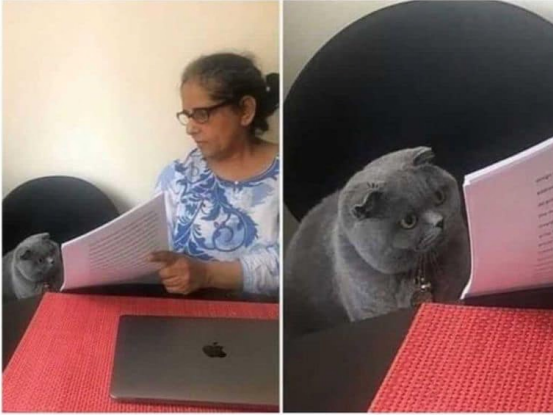


Session- 5

Deploy ON ECS

June 03, 2026

My manager showing me the code I wrote that took down the whole website



When you forget to **turn off** your AWS instance and receive your AWS bill



AGENDA

Brief Overview of AWS

To get our hands dirty → Deploy on ECS

Other Cloud Computing Platforms

1. AWS - Amazon
2. GCP - Google
3. Azure - Microsoft
4. Digital Ocean
5. Heroku
6. IBM
7. Oracle

AWS provides → Raw Infra → EC2
Pre-configured → ECS
Platform

AWS Services

1. EC2 -> Elastic Compute Cloud (Blank/Un-configured online-server)
2. S3 -> Simple Storage Service - Online Equivalent of Google Drive
3. ECS -> Elastic Container Service
4. ECR -> Elastic Container Registry (AWS Equivalent for Dockerhub)
5. SageMaker -> Google Colab for Amazon inside AWS.
6. EMR -> Distributed Computing - Apache Spark
7. IAM -> Identity and Management - Controls access throughout entire AWS System

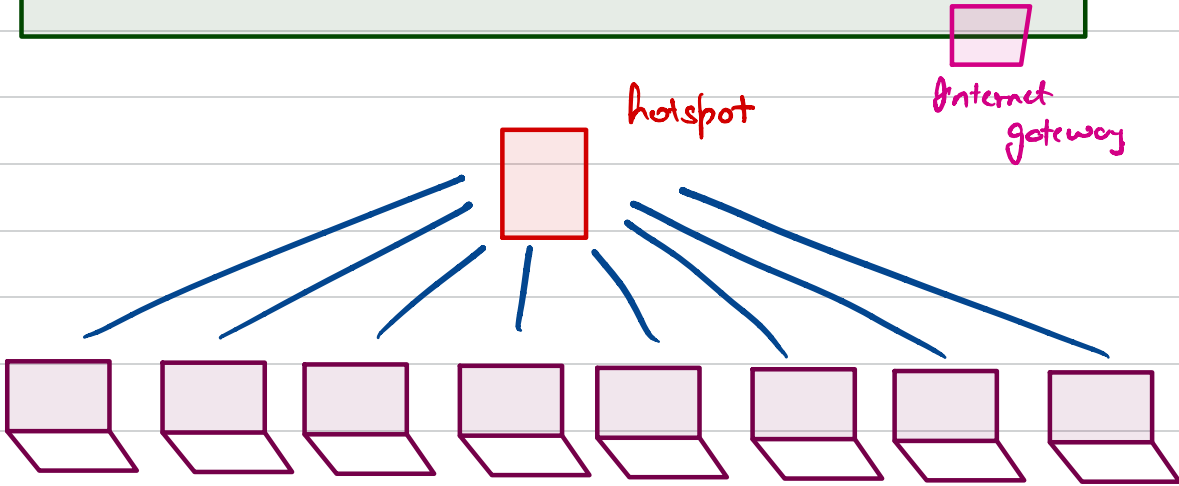
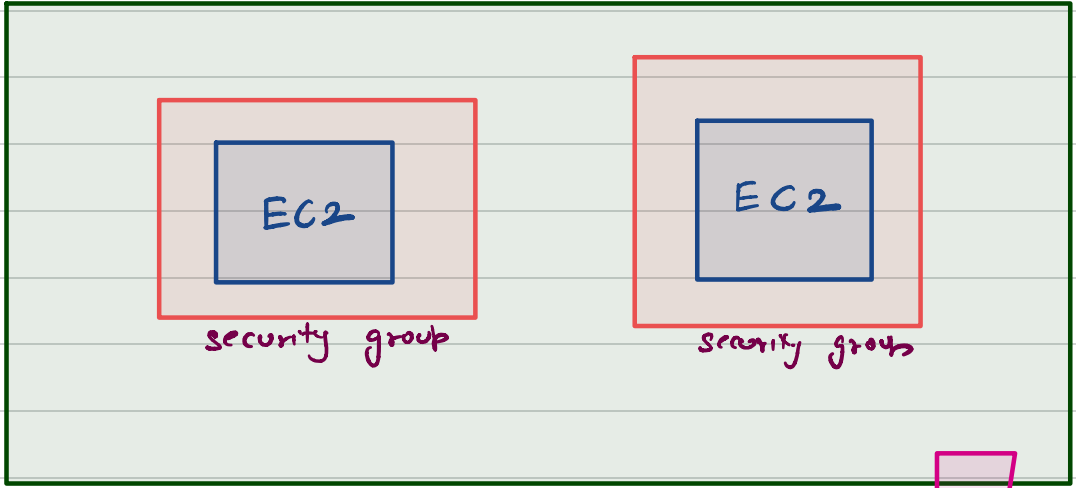
General Purpose	Compute Optimised	Memory Optimised	Accelerated Computing	Storage Optimised
 A1 ARM based core and custom silicon	 C4 Compute - CPU intensive apps and DBs	 R4 RAM - Memory Intensive apps and DB's	 P2 Processing optimised - Machine Learning	 H1 High Disk Throughput - Big data clusters
 T2 Tiny - Web servers and small DBs		 X1 Xtreme RAM - For SAP/Spark	 G3 Graphics Intensive - Video and Streaming	 I3 IOPS - No SQL DBs
 M4 Main - App servers and general purpose		 z1d High Compute and High Memory - Gaming	 F1 Field Programmable - Hardware acceleration	 D2 Dense Storage - Data Warehousing



EC2 instance types

	General Purpose		Compute Optimized	Memory Optimized		Accelerated Computing	Storage Optimized		
Type	t2	m5	c5	r4	x1e	p3	h1	i3	d2
Description	Burstable, good for changing workloads	Balanced, good for consistent workloads	High ratio of compute to memory	Good for in-memory databases	Good for full in-memory applications	Good for graphics processing and other GPU uses	HDD backed, balance of compute and memory	SDD backed, balance of compute and memory	Highest disk ratio
Mnemonic	t is for tiny or turbo	m is for main or happy medium	c is for compute	r is for RAM	x is for xtreme	p is for pictures	h is for HDD	i is for IOPS	d is for dense

VPC → Virtual Private Cloud



EC2

- ① Rent a server
For long time

ECS

- ① Pay as you go
- ② Auto scaling.

Spain → Ronda → Flat → 50K / month
Hotel → 5K / day

Auto scaling

- ① Horizontal Auto scaling.



12 GB RAM
512 GB SSD

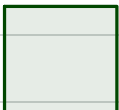


12 GB RAM
512 GB SSD

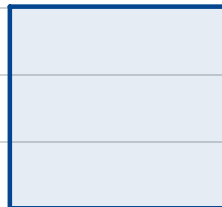


12 GB RAM
512 GB SSD

- ② VERTICAL Auto scaling

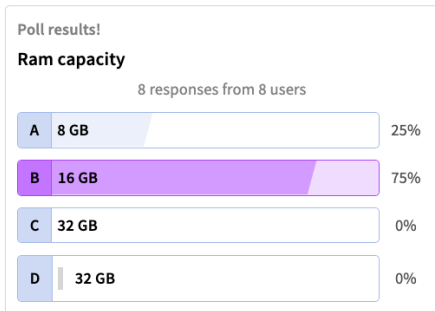
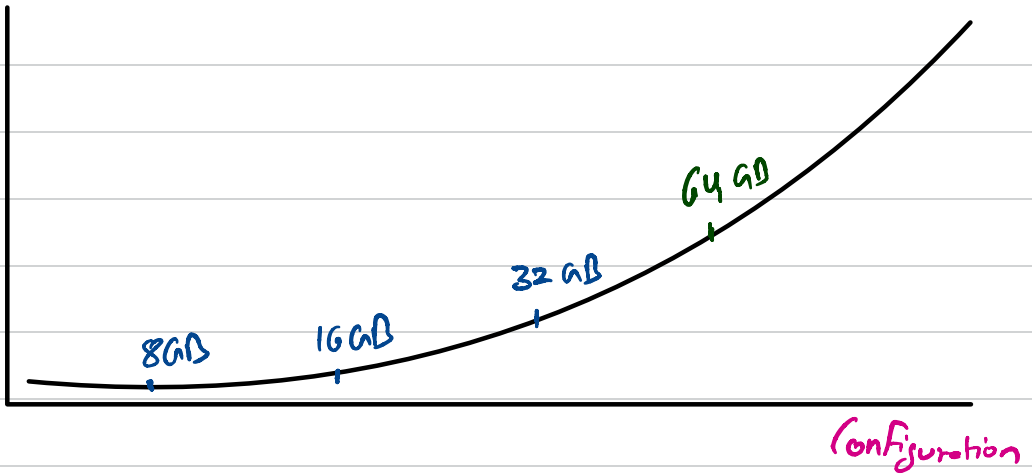


12 GB RAM
512 GB SSD



64 GB RAM
2TB SSD

Cost



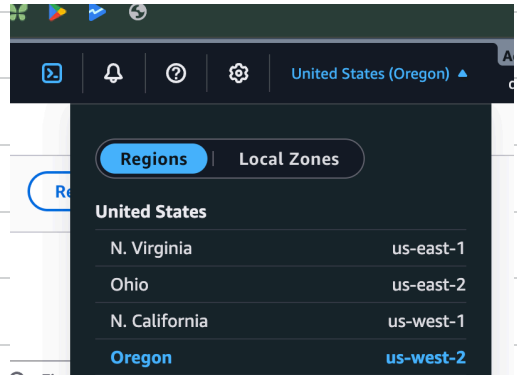
512 GB → 11 Ultra
8 stud.
11 2Kty

Steps for Model Deployment

1. Container-ize the model.
2. Push container to registry.
3. Define the specs of computer where to run it.
4. Setup Platform (VM, EC2)
5. Deploy and configure Network security.
6. Monitor

Same steps in context of Amazon ECS

1. Authenticate our computer with AWS, using IAM.
2. Container-ize our image, and push to ECR.
3. Create Cluster and task-definition.
4. Deploy and configure security-groups.

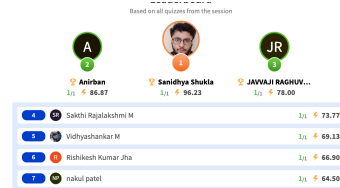


Which AWS service is responsible for managing containers and enabling developers to run applications without configuring the environment?

0 users have participated

A	Amazon Elastic Container Registry (Amazon ECR)	0%
B	Amazon Elastic Compute Cloud (Amazon EC2)	0%
☒	C Amazon Elastic Container Service (ECS)	0%
D	Amazon Simple Storage Service (Amazon S3)	0%

[End Quiz Now](#)

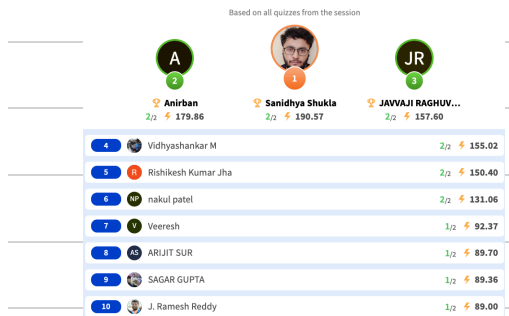


What does the term "repository" signify in the context of Amazon ECR?

0 users have participated

A	A server hosting web applications	0%
☒	B A storage location for Docker or OCI images	0%
C	A database for storing application data	0%
D	A collection of virtual machines	0%

[End Quiz Now](#)



What purpose does Amazon Elastic Container Registry (Amazon ECR) primarily serve in the AWS ecosystem?

0 users have participated

A

Managing serverless functions

0%

B

Configuring virtual machines

0%

C

Controlling access to S3 buckets

0%

✓

D

Storing and managing container images securely

0%

End Quiz Now

A

2

Anirban

3/3

268.70

I

1

Sanidhya Shukla

3/3

289.10

JR

3

JAVVAJI RAGHUV...

3/3

237.39

4

Vidhyashankar M

3/3

236.54

5

Veeresh

2/3

184.93

6

ARIJIT SUR

2/3

180.57

7

SAGAR GUPTA

2/3

179.87

8

J. Ramesh Reddy

2/3

178.03

9

Kuntal Kole

2/3

174.10

10

Prashant Barkale

2/3

165.60

What is the primary function of a virtual private cloud (VPC) in AWS?

0 users have participated

A

Storing and managing virtual machine images

0%

✓

B

Providing a dedicated virtual network for an AWS account

0%

C

Optimizing database queries

0%

D

Enabling high-performance computing

0%

End Quiz Now

A

2

Anirban

4/4

361.60

I

1

Sanidhya Shukla

4/4

387.00

V

3

Veeresh

3/4

276.76

4

SAGAR GUPTA

3/4

274.60

5

ARIJIT SUR

3/4

268.77

6

J. Ramesh Reddy

3/4

267.79

7

Prashant Barkale

3/4

242.26

8

JAVVAJI RAGHUVVEER

3/4

237.39

9

Vidhyashankar M

3/4

236.54

10

Kuntal Kole

2/4

174.10